

Darren Wang

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EDUCATION

University of Pennsylvania

B.S.E. in Computer Science; GPA: 3.99/4.0

May 2028

Philadelphia, PA

- **Relevant Coursework:** Operating Systems, Databases, Algorithms, Machine Learning, Computer Vision, Big Data Analytics, Scalable Cloud Computing, Computer Systems, Graph Neural Networks, Natural Language Processing.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, OCaml, SQL, JavaScript, HTML/CSS

ML/Data: PyTorch, scikit-learn, NumPy, polars, pandas, LightGBM, FinBERT, Plotly, Apache Spark

Tools: Docker, AWS (EC2, EMR, RDS, DynamoDB), PostgreSQL, MongoDB, Neo4j/Cypher, Node.js, React

EXPERIENCE

Certitude (YC Backed)

Feb. 2025 – May 2025

Software Engineering Intern

Philadelphia, PA

- Built vector embeddings to power semantic search and content discovery for a full-stack EdTech platform serving 10,000+ users.
- Developed a Python scraping and data cleaning pipeline for large-scale datasets used in AI-powered personalization features.

Penn Spark

Feb. 2026 – Present

Software Developer – Atelic Travel Client Project

Philadelphia, PA

- Rebuilt public-facing marketing pages for Atelic, a travel-planning startup, using React, Tailwind CSS, and React Router.
- Built reusable navigation, landing, footer, and contact components with responsive layouts, App Store CTAs, custom typography, and Formspree-backed lead capture.

University of Pittsburgh

Aug. 2023 – Present

Research Intern

Pittsburgh, PA

- Discovered significant correlation ($r \approx 0.6$) between AlphaFold 3 confidence scores and Gibbs free energy (ΔG).
- Engineered a protocol utilizing ANI-2x and a custom conjugate gradient optimizer to rerank AlphaFold 2 predictions, achieving 34% Top-1 accuracy on challenging blind test sets.

Penn Aerospace Club

Sept. 2024 – Present

Software Engineer

Philadelphia, PA

- Developed low-level C++ firmware for microcontrollers to process real-time atmospheric and navigational sensor data.
- Built machine learning models to predict weather patterns using data collected from semesterly high-altitude balloon launches.

PROJECTS

Social Sentiment Trading Study | *Python, polars, FinBERT, scikit-learn*

May 2026

- Built a financial sentiment modeling pipeline combining Bluesky posts with Yahoo Finance hourly OHLCV data to study short-horizon stock movement prediction.
- Engineered lagged return, rolling volume, sentiment, posting-activity, and time-based features, producing a 13K-row, 70-feature ticker-hour dataset.
- Benchmarked Logistic Regression, Random Forest, and Histogram Gradient Boosting models with chronological splits, finding limited out-of-sample signal with ROC-AUC near random.

Predictive VM Resource Allocator | *Python, scikit-learn, PyTorch*

Nov. 2025

- Designed a machine learning pipeline to predict cloud VM criticality using 30-day historical usage traces and aggregated subscription features.
- Benchmarked 12 architectures, achieving 0.83 AUC-ROC with LightGBM and 79.3% accuracy with a Pruned Random Forest model.

Generative 3D Collision Renderer | *Python, Blender API, Computer Vision*

Dec. 2025

- Deployed a full-stack web application that processes user-uploaded 2D images into textured 3D meshes via TripoSR inference.
- Engineered a Blender API rendering system that predicts physical material properties and automates physics-based collision simulations.

LEADERSHIP

Steel City Codes

Oct. 2022 – May 2024

Chapter Founder & Director

Allison Park, PA

- Founded the local chapter and led 6 instructors in teaching Python curriculum to ~130 elementary students.